Intergenerational Housing Misallocation and Fertility
Compact Theory and Quantitative Lifecycle Model

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Abstract

Children require goods and housing space. The space requirement matters because family-sized housing is harder to access than small rental housing: large units are concentrated in ownership, ownership requires down payments, and tax rules can induce older owners to retain houses whose marginal value to them is below the value to constrained young families. The paper has two parts. Part I gives a compact two-period model that isolates the mechanism and delivers goods-equivalent wedges for young access constraints, old-owner lock-in, and fertility. The compact model also makes clear which aggregate birth effects come from completed fertility among entrants, which come from entry into the housing market, and which come from tenure switching. Part II embeds the same mechanism in a finite-life quantitative lifecycle model with income risk, tenure and house-size choice, down-payment constraints, transaction costs, bequests, child aging, and a single aggregate housing-services market.

1 Introduction

A child changes a household's housing problem. It is not only another set of goods to buy, or more time to spend at home. It is often a need for a different dwelling: another bedroom, a larger living space, or a unit where raising a child is feasible. In many high-cost cities this margin is hard to adjust. Large rental units are scarce, family-sized homes are concentrated in ownership, and ownership requires down payments and borrowing capacity at ages when many households have little wealth.

This paper studies whether those space and financial constraints affect final family size and the number of births generated by a cohort. That distinction matters. A housing shock can move births across years without changing completed fertility. It can also change local births by changing who enters a high-cost housing market, even if completed fertility among incumbent households moves little. The question here is when housing constraints change completed fertility, when they mainly change entry and local scale, and how both effects depend on who can access family-sized space over the lifecycle.

The mechanism has three parts. First, children raise the household's demand for space. If the rental market does not offer sufficiently large units, the shadow price of an additional room can exceed the posted rent. Second, the natural substitute is ownership, but buying a family-sized home requires liquidity and credit. A young household may value the larger unit and still be unable to purchase it. Third, older owners may remain in large homes even when their own marginal value of space is low, because transaction costs and tax rules subsidize staying put. Housing scarcity then shows up not only as a high price of space, but also as a mismatch between who occupies family-sized units and who values them for fertility.

I develop the argument in two steps. The first part of the paper gives a compact two-period model that isolates the mechanism. It expresses the rental space constraint, the ownership finance constraint, old-owner lock-in, and the fertility distortion as goods-equivalent wedges. The second part embeds the same economics in a finite-life quantitative model with income risk, tenure and house-size choice, down-payment constraints, transaction costs, bequests, child aging, and a single aggregate housing-services market. The model combines lifecycle fertility choice (Sommer, 2016; Doepke and Kindermann, 2019) with housing tenure choice, collateral constraints, and owner-occupied housing (Henderson and Ioannides, 1983; Sommer et al., 2013; Sommer and Sullivan, 2018). The compact housing block follows the one-market floorspace logic in Coven et al. (2025): all housing services clear in one market, while tenure, borrowing, and tax rules determine who can actually occupy family-sized units.

2 Related Literature

This paper contributes to three strands of literature.

First, it contributes to work on fertility decline, fertility choice, and the housing determinants of fertility. Since Becker (1960) and Becker and Lewis (1973), economics has treated children as choice variables that enter preferences and require resources. Quantitative versions of this approach put that choice into a lifecycle environment with income risk, savings, timing, and intrahousehold bargaining (Jones et al., 2010; Sommer, 2016; Doepke and Kindermann, 2019). Recent work on the U.S. fertility decline emphasizes that falling birth rates are hard to explain with a single postponement margin or one obvious period shock (Kearney et al., 2022). The housing literature shows that housing prices, housing wealth, and mortgage finance move fertility behavior: Dettling and Kearney (2014) find opposite house-price effects for owners and nonowners, Lovenheim and Mumford (2013) find that housing wealth raises homeowners' fertility, Cumming and Dettling (2024) show that mortgage-rate pass-through affects births through cash flows, and Hacamo (2021) finds that mortgage deregulation increased home purchases and childbearing for young households. The structural housing-fertility literature studies the same broad connection in equilibrium: Couillard (2025) models joint location, house-size, and fertility choice, and shows that the supply of large units matters for aggregate fertility.

The object in this paper is closer to completed fertility than to a current birth rate, so I separate timing evidence from final-family-size evidence within this same strand. Simon and Tamura (2009) use U.S. Census data from 1940–2000 and find that higher rents per room are associated with lower fertility, including in completed-fertility specifications for older households. Clark (2012) is an important caution: expensive U.S. housing markets delay first births by several years, but do not clearly reduce completed fertility. More recent evidence is more favorable to the family-size interpretation. Japaridze and Sayour (2024) link unaffordable homeownership to delayed entry into motherhood and lower completed fertility. Dettling and Kearney (2025) study the FHA and VA mortgage programs and find that low-down-payment, long-term mortgage access increased births during the U.S. baby boom; their Census evidence on children ever born is consistent with higher completed fertility for groups with access to the programs. van Doornik et al. (2025) exploit randomized Brazilian housing-credit lotteries and find larger lifetime fertility effects when housing access arrives earlier in the childbearing years. I use this literature to discipline the distinction between three objects: current births, completed fertility among entrants, and births generated by a cohort after entry into the housing market is allowed to respond.[1]

[1] The same issue appears in demography and life-course research, which documents that fertility, homeownership,

Second, the paper contributes to the literature on intergenerational housing allocation and lock-in. The closest paper is Coven et al. (2025). Their model studies how property taxes, capitalization, and financial constraints allocate housing between older homeowners and younger constrained households. This paper uses the same core allocation problem: young households may value family-sized space but face down-payment constraints, while older incumbents may retain housing that would be valuable to younger families. The departure is that I make children part of the household problem. Reallocating family-sized space is then not only a homeownership or welfare issue; it also changes the price of children, completed fertility among entrants, and local births through entry.

The lock-in mechanism is also related to work on mortgage lock-in and housing ladders. Fonseca and Liu (2023) show that locked-in mortgage rates reduce mobility and dampen labor reallocation. In a spatial housing-ladder model, Fonseca et al. (2026) show that lock-in reduces moves down the ladder and keeps missing downsizers in larger homes, with equilibrium effects on prices and rents. My lock-in force comes from transaction costs and the tax treatment of capital gains at death rather than from fixed-rate mortgages, but the allocation logic is the same: incumbent frictions keep housing with households whose current marginal value of space may be low, tightening access for younger households whose family choices are space-intensive.

Third, the paper contributes to structural work on housing, tenure choice, and spatial allocation. The housing block builds on models in which tenure, collateral constraints, and owner-occupied housing shape household behavior (Henderson and Ioannides, 1983; Sommer et al., 2013; Sommer and Sullivan, 2018). It also relates to spatial-equilibrium work in which housing supply and local prices allocate households across space (Rosen, 1979; Roback, 1982; Gyourko et al., 2013; Hsieh and Moretti, 2019). The compact model keeps one floorspace market and uses sufficient statistics for the allocation wedges. The quantitative model then adds lifecycle income risk, tenure and house-size choice, children aging out of the household, bequests, and housing supply. The aim is to connect the intergenerational allocation margins in the housing literature to the fertility objects emphasized in the first strand.

Part I

Compact Analytical Model

3 Environment

Agents. The economy is populated by overlapping generations of young and old households. A young household has type

$$i = (y_i, a_i),$$

where $y_i > 0$ is disposable income and $a_i \geq 0$ is entry wealth, inherited from the previous old generation. Conditional on entry, the young household chooses tenure, housing, completed fertility, and saving. Old households are indexed by j. An old incumbent is last period's young owner and enters old age with financial wealth, the house she bought, and an accrued taxable gain. She chooses

dwelling size, and residential transitions move together (Mulder and Billari, 2010; Ström, 2010; Kulu and Steele, 2013; Chudnovskaya, 2019; Tocchioni et al., 2021). Demographic accounts of low fertility also emphasize broader changes in family formation, marriage, and living arrangements (Myrskylä et al., 2013; Lesthaeghe, 2014). I use that evidence as motivation for housing as a family-formation input, while the model below treats fertility as an economic choice over the lifecycle.

consumption, how much housing to retain, and the estate she leaves at exit. An old renter carries no housing and no capital-gains wedge.

Masses and demography. At the start of period t, a mass M_t of potential young households is drawn from a distribution $G_{Y,t}$. A potential young household either enters the modeled housing market or takes an outside option. Nonentrants live outside the modeled market in both periods. Current young entrants become next period's old households, and the next potential-young cohort is formed by births plus a renewal inflow R_t. The stationary cohort mass is denoted by $\bar{M}$.

Preferences. Children raise both consumption needs and housing needs. In the compact model fertility is continuous. A young household choosing consumption c_i, housing h_i, and fertility n_i has utility

$$u_i^Y(c_i, h_i, n_i) = \log(c_i - \chi n_i) + \alpha \log(h_i - \kappa n_i) + \vartheta \log n_i, \qquad \alpha, \vartheta, \chi, \kappa > 0. \tag{1}$$

The term χn_i is the goods requirement of children and κn_i is their housing-space requirement. A marginal child therefore uses the same housing stock as the parents.

Old households value consumption, housing, and bequests:

$$u^O(c_j^O, h_j^O, e_j) = \log c_j^O + \gamma \log h_j^O + \mathcal{B}(e_j), \qquad \gamma > 0, \tag{2}$$

where $e_j \geq 0$ is the estate and $\mathcal{B}$ is increasing and concave, possibly identically zero. Bequest motives and true old housing utility are preferences; they are not wedges.

Housing. There is a fixed stock $\bar{H}$ of divisible floorspace. Households occupy floorspace by renting or owning. The two tenure modes have different size limits:

$$\mathcal{H}^R = \{h : 0 < h \leq \bar{h}^R\}, \qquad \mathcal{H}^O = \{h : 0 < h \leq \bar{h}^O\}, \qquad 0 < \bar{h}^R < \bar{h}^O. \tag{3}$$

This is the family-housing segmentation: small units are available to rent, but large family-capable units are mainly reached through ownership. All floorspace trades in one market. The user cost of housing is q and the asset price is P, linked by

$$q = \rho P, \qquad \rho = r + \delta + \tau^p, \tag{4}$$

where r is the interest rate, δ is depreciation, and τ^p is the property-tax rate. For a given user cost q, a higher property tax lowers the asset price P and therefore lowers the down payment needed to buy a given house.

Financing. A buyer can finance at most share $\phi \in (0, 1)$ of the purchase price. Buying h units therefore requires the cash down payment

$$(1 - \phi)Ph. \tag{5}$$

Renters cannot borrow. Households can save in a bond at return $1 + r$. The compact model treats the down-payment constraint as an access constraint: it limits the amount of owner housing a young household can occupy.

Capital-gains taxation and step-up basis. Selling a house triggers a capital-gains tax. Dying with the house gives a stepped-up basis, so the accrued gain is forgiven. In a stationary real model, the gain comes from nominal price growth. If dollar prices grow at rate ϖ, a house bought one period ago has a taxable paper gain even when its real value is unchanged. The real flow-equivalent tax wedge per unit sold is

$$\bar{\ell} = \tau^g \frac{P\varpi}{1 + \varpi}, \tag{6}$$

after normalizing the period conversion factor to one. The wedge is not a preference and not a resource cost. It is a private tax liability avoided by holding the house until death.

4 Households

Conditional on entry, young household i chooses a tenure mode $m \in \{R, O\}$, housing, fertility, consumption, and savings. For $m \in \{R, O\}$,

$$\begin{aligned} W_i^m(q) = \max_{c_i, h_i, n_i, a_i'} \quad & \log(c_i - \chi n_i) + \alpha \log(h_i - \kappa n_i) + \vartheta \log n_i \\ & + \beta \left[\mathbf{1}\{m = R\} V^R(a_{j'}) + \mathbf{1}\{m = O\} V^O(a_{j'}, h_i) \right] \end{aligned} \tag{7}$$

subject to

$$c_i + qh_i + a_i' = y_i + a_i, \tag{8}$$

$$h_i \in \mathcal{H}^m, \qquad a_i' \geq 0,$$

$$\mathbf{1}\{m = O\}(1 - \phi)Ph_i \leq a_i, \tag{9}$$

$$c_i - \chi n_i > 0, \qquad h_i - \kappa n_i > 0.$$

Using $P = q/\rho$, the largest owner house feasible for type i is

$$H_i^O(q) = \min \left\{ \bar{h}^O, \frac{a_i \rho}{(1 - \phi)q} \right\}, \qquad H_i^R = \bar{h}^R. \tag{10}$$

H_i^m is the effective family-housing cap. A renter is capped by the rental stock. A buyer is capped by the owner size limit or by collateral, whichever is shorter.

An old incumbent enters with financial wealth a_j, carried housing H_j^0, and an accrued gain. She chooses consumption, retained housing, and an estate:

$$\begin{aligned} V^O(a_j, H_j^0) = & \max_{c_j^O, h_j^O, e_j} \; u^O(c_j^O, h_j^O, e_j) \\ & \text{s.t.} \quad c_j^O + e_j + qh_j^O = a_j + qH_j^0 - \bar{\ell}(H_j^0 - h_j^O), \qquad 0 \leq h_j^O \leq H_j^0. \end{aligned} \tag{11}$$

Selling one unit nets $q - \bar{\ell}$, so stepped-up basis acts as a subsidy to retaining the house. An old renter has no carried housing and no tax:

$$V^R(a_j) = \max_{c_j^O, h_j^O, e_j} u^O(c_j^O, h_j^O, e_j) \quad \text{s.t.} \quad c_j^O + e_j + qh_j^O = a_j, \qquad 0 < h_j^O \leq \bar{h}^R. \tag{12}$$

5 Entry and Equilibrium

A potential young household can enter the housing market or take an outside option with value $\bar{W}^E$. Entry has Type-I extreme-value shocks with scale κ_E. The inside value is

$$W_i^Y(q) = \max\{W_i^R(q), W_i^O(q)\},$$

and the entry probability is

$$\pi_i^E(q) = \frac{\exp\{W_i^Y(q)/\kappa_E\}}{\exp\{W_i^Y(q)/\kappa_E\} + \exp\{\bar{W}^E/\kappa_E\}}. \tag{13}$$

Let $n_i^Y(q)$ be fertility in the inside tenure chosen by type i, and let $\bar{n}_i^E$ be fertility in the outside option. Births generated by the potential-young cohort are

$$N(q) = \bar{M} \int \left[\pi_i^E(q) n_i^Y(q) + (1 - \pi_i^E(q)) \bar{n}_i^E \right] \mathrm{d}G_Y(i). \tag{14}$$

The baseline normalization sets $\bar{n}_i^E = 0$, but the entry-margin terms below are written so that outside fertility can be restored by replacing inside fertility with the gap $n_i^Y - \bar{n}_i^E$. In a stationary demographic equilibrium, births and renewal close the young cohort:

$$N + \bar{R} = \bar{M}. \tag{15}$$

The local formulas below hold $(\bar{M}, G_Y, F_O, F_R, \bar{R})$ fixed. A full demographic counterfactual instead recomputes the stationary fixed point, or else states which object is held fixed and which object absorbs the birth change.

Let F_O and F_R be the stationary measures of old incumbents and old renters. Aggregate floorspace demand is

$$H^D(q) = \bar{M} \int \pi_i^E(q) h_i(q)\, \mathrm{d}G_Y(i) + \int h_j^O(q)\, \mathrm{d}F_O(j) + \int h_j^O(q)\, \mathrm{d}F_R(j). \tag{16}$$

The compact equilibrium clears the single stock:

$$H^D(q) = \bar{H}, \qquad P = q/\rho. \tag{17}$$

Stationarity requires current young owners to become next period's old incumbents, current young renters to become old renters, and births plus the renewal inflow to reproduce the next young cohort. The transition rules assign old estates to entry wealth and are otherwise left general; the local results below condition on the stationary cross-section.

6 Equilibrium Wedges

The model's sufficient statistics are marginal values of space in consumption goods. Let λ_i^m be the multiplier on the young household's budget, θ_i^m the multiplier on the size limit, and μ_i the multiplier on the owner's down-payment constraint. For renters,

$$\frac{\alpha(c_i^R - \chi n_i^R)}{h_i^R - \kappa n_i^R} = q + \zeta_i^R, \qquad \zeta_i^R = \frac{\theta_i^R}{\lambda_i^R} \geq 0. \tag{18}$$

A capped renter values one more unit of housing above the rent it would command.

For young owners, anticipation of the future capital-gains liability raises the effective price of owning family-sized space. Under interior saving and interior old retention,

$$q^O = q + \frac{\bar{\ell}}{1 + r}. \tag{19}$$

The owner access gap is

$$\zeta_i^O = \underbrace{\frac{\mu_i}{\lambda_i^O}(1 - \phi)P}_{\zeta_i^{O,F}\text{, down-payment component}} + \underbrace{\frac{\theta_i^O}{\lambda_i^O}}_{\text{owner-size component}} \geq 0, \tag{20}$$

so the owner's marginal value of space is

$$\frac{\alpha(c_i^O - \chi n_i^O)}{h_i^O - \kappa n_i^O} = q^O + \zeta_i^O. \tag{21}$$

When the owner size limit is slack, $\zeta_i^O = \zeta_i^{O,F}$ and the gap is a financing gap.

The old incumbent's interior retention condition is

$$\frac{\gamma c_j^O}{h_j^O} = q - \bar{\ell}. \tag{22}$$

She keeps marginal space she values below the market user cost because selling it realizes the taxable gain. An old renter has no such wedge and, at an interior choice, values space at q.

The fertility condition prices the marginal child in goods. In tenure mode m, with $q^R = q$ and q^O given by (19),

$$\underbrace{\frac{\vartheta(c_i^m - \chi n_i^m)}{n_i^m}}_{\text{willingness to pay for a marginal child}} = \underbrace{\chi + \kappa(q^m + \zeta_i^m)}_{\text{full price of a child}}. \tag{23}$$

A child's space requirement is priced at the household's own shadow value of space. A binding access constraint therefore acts as an implicit child tax $\kappa\zeta_i^m$.

7 Efficiency and Fertility

The planner comparison holds the stationary cross-section, entry probabilities, tenure assignments, savings, and real size-limit constraints fixed at the instant of the variation. The planner counts true household preferences and true resource costs. It does not count the avoided capital-gains tax as a social benefit.

Proposition 1 (Local young-old housing misallocation)**.** *Consider a stationary equilibrium. Hold entry, tenure assignments, savings, and real size-limit constraints fixed. Take a young owner whose owner size limit is slack and whose financing constraint binds, so $\zeta_i^{O,F} > 0$, and an old incumbent at an interior retention choice. If one unit of existing housing can be reassigned from the old incumbent to the young owner at zero real implementation cost, the planner's goods-equivalent surplus is*

$$\underbrace{(q^O + \zeta_i^{O,F})}_{\textit{young buyer's marginal value}} - \underbrace{(q - \bar{\ell})}_{\textit{old incumbent's marginal value}} = \zeta_i^{O,F} + \frac{\bar{\ell}}{1+r} + \bar{\ell}. \tag{24}$$

If this expression is positive, the competitive allocation does not solve the planner's conditional allocation problem.

The market price cancels. What remains are the young household's financing gap, the buyer's anticipated gains tax, and the old incumbent's realized gains tax. The two tax terms are transfers, not resource costs. If implementing the reassignment requires real cost r_i^F per unit, that cost subtracts from the surplus. If the down-payment constraint is a primitive technological enforcement constraint and no instrument can relax it, the variation is not feasible.

Old occupancy is not inefficient by itself. An old owner who keeps a large house because she values it, or because she values leaving bequests, is using housing for reasons the planner also counts. The inefficiency is the part of retention supported by the forgiveness of capital-gains taxes at death, paired with young households who value family-sized space above its market user cost.

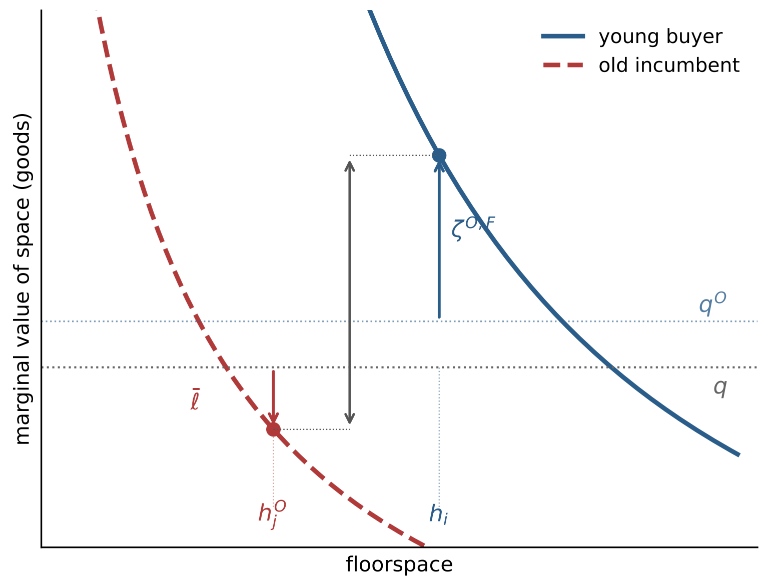

Figure 1: Local misallocation. A collateral-capped young buyer values marginal space at $q^O + \zeta^{O,F}$, while a locked-in old incumbent values retained space at $q - \bar{\ell}$. The vertical distance is the goods-equivalent surplus from a compensated marginal reassignment.

The same cap determines a fertility response. Fix a tenure mode and a household whose housing is pinned at the effective family-housing cap H. Concentrating the savings rule into the budget, fertility along the binding cap solves

$$\frac{\vartheta}{n} = \frac{(1+k)\chi}{w - q^m H - \chi n} + \frac{\alpha\kappa}{H - \kappa n},$$

where $w = y + a$ and k is the savings load on adult consumption. Differentiating with respect to the cap gives

$$\frac{\mathrm{d}n}{\mathrm{d}H} = \frac{\dfrac{\alpha\kappa}{(H-\kappa n)^2} - \dfrac{(1+k)\chi q^m}{(w - q^m H - \chi n)^2}}{\dfrac{\vartheta}{n^2} + \dfrac{(1+k)\chi^2}{(w - q^m H - \chi n)^2} + \dfrac{\alpha\kappa^2}{(H-\kappa n)^2}}. \tag{25}$$

Relaxing access raises fertility when the space-relief force dominates the budget-tightening force. A useful sufficient condition along every binding cap is

$$\chi \leq \frac{(1+k)\kappa q^m}{\alpha}. \tag{26}$$

The derivative connects the misallocation wedge to fertility: policies that raise the effective family-housing cap matter only for households whose binding constraint they actually relax.

This household derivative aggregates into a local sufficient statistic for births. Let z be a policy parameter. For renters, $H_i^R = \bar{h}^R$; for owners,

$$H_i^O(q) = \min\left\{\bar{h}^O, \frac{a_i \rho}{(1-\phi)q}\right\}.$$

Let $\mathcal{E}_m(z)$ be the set of types for whom tenure mode m is the unique active tenure, the effective cap H_i^m is strictly binding, and the binding component of the cap is locally unique. The pass-through of policy z to the effective family-housing cap is

$$\mathcal{P}_i^m(z) = \frac{\partial H_i^m(z)}{\partial z}.$$

For an exposed type, write residual adult consumption and residual space as

$$x_i^m = c_i^m - \chi n_i^m, \qquad s_i^m = h_i^m - \kappa n_i^m.$$

The fixed-mode fertility response to one more unit of permitted family housing is

$$\Psi_i^m = \frac{\dfrac{\alpha\kappa}{(s_i^m)^2} - \dfrac{\chi q^m}{(1+k)(x_i^m)^2}}{\dfrac{\vartheta}{(n_i^m)^2} + \dfrac{\chi^2}{(1+k)(x_i^m)^2} + \dfrac{\alpha\kappa^2}{(s_i^m)^2}}. \tag{27}$$

The value of relaxing the cap in utility units is

$$\Theta_i^m \equiv \frac{\partial W_i^m}{\partial H_i^m} = \frac{\alpha}{s_i^m} - \frac{q^m}{x_i^m} = \frac{\zeta_i^m}{x_i^m}. \tag{28}$$

Thus the same access gap that measures misallocation also governs the entry response.

Proposition 2 (Local fixed-state fertility statistic)**.** *Holding prices, tenure choices, active sets, outside fertility, and the stationary demographic state fixed, the local effect of a cap-shifting policy on cohort births is*

$$\left.\frac{\mathrm{d}N}{\mathrm{d}z}\right|_{q,\text{tenure},\mathcal{A}} = \bar{M} \sum_{m\in\{R,O\}} \int_{\mathcal{E}_m(z)} \left[\pi_i^E \Psi_i^m + (n_i^m - \bar{n}_i^E)\frac{\pi_i^E(1-\pi_i^E)}{\kappa_E}\Theta_i^m\right] \mathcal{P}_i^m(z)\,\mathrm{d}G_Y(i). \tag{29}$$

The first term is the intensive margin: already-entered households change completed fertility by Ψ_i^m for each unit of cap relief. The second term is the entry margin: cap relief raises the value of entering, and marginal entrants bring their inside fertility net of the fertility they would have had outside the modeled market. Both terms are gated by the same exposure set. A policy aimed at a slack constraint has zero pass-through in this channel.

At one benchmark point the fertility response and the access gap are the same object.

Corollary 1 (Equal scaling: one statistic for both margins)**.** *Suppose $\chi = (1+k)\kappa q^m/\alpha$, equal scaling at lifetime prices in mode m, and the effective cap binds strictly. Then*

$$\Psi_i^m = \frac{\kappa}{\alpha}\frac{\zeta_i^m}{x_i^m}\,\omega_i^m, \qquad \omega_i^m = \frac{\dfrac{\alpha}{s_i^m} + \dfrac{q^m}{x_i^m}}{\dfrac{\vartheta}{(n_i^m)^2} + \dfrac{\chi^2}{(1+k)(x_i^m)^2} + \dfrac{\alpha\kappa^2}{(s_i^m)^2}} > 0. \tag{30}$$

Under equal scaling, the normalized access gap ζ_i^m/x_i^m is simultaneously the misallocation wedge of the efficiency proposition, the fertility exposure Ψ_i^m up to the positive weight ω_i^m, and the entry response Θ_i^m. This is the object the quantitative model measures: Part II reports the distribution of ζ among fertile households and the share of the fertility-response mass that sits on $\zeta > 0$ types.

One pass-through has a closed form that Part II uses. For a collateral-capped owner, $H_i^O = a_i\rho/[(1-\phi)q]$ with $\rho = r + \delta + \tau^p$. Holding the user cost fixed, a property-tax increase raises ρ, lowers the asset price $P = q/\rho$, and thereby lowers the down payment per unit of house:

$$\left.\frac{\partial H_i^O}{\partial \tau^p}\right|_q = \frac{a_i}{(1-\phi)q} > 0. \tag{31}$$

A property tax, capitalized into the asset price, relaxes the collateral cap by $a_i/[(1-\phi)q]$ units per point of tax, and the birth response follows from Proposition 2 integrated over the collateral-exposed set. Whether the relief creates births in practice depends on whether exposed households are near their fertility margin when the relief arrives. Part II shows that timing, not exposure, is the binding failure.

The formula is fixed-tenure accounting. If tenure choice is deterministic and the type distribution has density near the rent-own boundary, a policy can also move households across that boundary. Let

$$\Delta_i(z) = W_i^O(z) - W_i^R(z), \qquad \mathcal{I}(z) = \{i : \Delta_i(z) = 0\},$$

and suppose $\nabla_i \Delta_i(z) \neq 0$ on $\mathcal{I}(z)$, with unique mode optima and no mass points at kinks. The tenure-switching component is

$$\left.\frac{\mathrm{d}N^{\text{switch}}}{\mathrm{d}z}\right|_{q,\mathcal{A}} = \bar{M} \int_{\mathcal{I}(z)} \pi_i^E \left(n_i^O - n_i^R\right) \frac{\partial_z \Delta_i(z)}{\|\nabla_i \Delta_i(z)\|} f(i)\,\mathrm{d}S(i), \tag{32}$$

where $f(i)$ is the type density and $\mathrm{d}S(i)$ is surface measure on the indifference set, oriented toward ownership. The total fixed-price effect is

$$\left.\frac{\mathrm{d}N}{\mathrm{d}z}\right|_q = \left.\frac{\mathrm{d}N}{\mathrm{d}z}\right|_{q,\text{tenure},\mathcal{A}} + \left.\frac{\mathrm{d}N^{\text{switch}}}{\mathrm{d}z}\right|_{q,\mathcal{A}}. \tag{33}$$

There is no separate entry-boundary term under logit entry because the inside value is continuous at the tenure boundary. The discontinuity is in conditional fertility, $\pi_i^E(n_i^O - n_i^R)$. If $q^O = q^R$, the two tenure modes differ only through caps and the boundary fertility jump disappears when the boundary allocation is the same in both modes. With a tax wedge, ownership is the more expensive way to access family space; the switching term is first order, but its sign depends on the fertility jump of boundary households.

The sign of the switching component depends on the instrument. Owner-access policies move boundary households toward the mode with the larger cap, so their switching births reinforce the intensive margin. Relaxing the rental cap works the other way: it makes renting better precisely for households near the rent-own boundary, and marginal owners pulled back into renting carry the lower owner-mode fertility jump with a negative sign. Rental-side instruments therefore carry a nonpositive switching component, a caveat that returns in Part II's rental-cap experiments.

Part II

Quantitative Lifecycle Model

8 Purpose and Timing

The quantitative model embeds the same mechanism in a finite-life economy. It has the standard lifecycle structure: households age, face income risk, save, choose whether to rent or own, choose housing size, decide fertility, and eventually leave bequests. The housing market has one aggregate user cost and one asset price. Tenure matters because it changes the size menu, financing constraints, transaction costs, and tax exposure.

Time is discrete. Households enter adult life at age $a = 1$, retire after age J_R, and die at the end of age J. The individual state at the beginning of a period is

$$x = (b, z, d, a, n, s). \tag{34}$$

Here b is liquid wealth, $z \in \mathcal{Z}$ is idiosyncratic productivity, d is tenure and housing position, a is age, n is completed fertility, and s is the child-at-home state. Tenure belongs to

$$\mathcal{D} = \{R, O_1, \ldots, O_K\}. \tag{35}$$

R denotes renting. O_k denotes ownership of a house with physical size H_k, where $H_1 < \cdots < H_K$. Renters choose housing continuously up to $\bar{h}^R$; owners choose from the discrete ladder. The ladder is the quantitative version of the family-housing segmentation in Part I.

Income for working households is

$$y(a, z) = (1 - \tau_{\text{pay}}) w e_a z, \qquad z' \sim \Pi^z(\cdot|z), \qquad a \leq J_R, \tag{36}$$

and retired households receive pension income p^{ret}. The wage is common apart from age and productivity because the quantitative mechanism is housing access, tenure, and lifecycle wealth.

9 Preferences and Bequests

Let $m(n, s)$ be the number of dependent children living at home. Completed fertility n and current dependents differ because children eventually mature. Household needs are

$$\begin{aligned} \bar{c}(n, s) &= \bar{c}_0 + \bar{c}_n m(n, s), \\ \bar{h}(n, s) &= \bar{h}_0 + \bar{h}_{\text{jump}} \mathbf{1}\{m(n, s) \geq 1\} + \bar{h}_n m(n, s). \end{aligned} \tag{37}$$

The first child can create a discrete housing-space jump; additional dependent children raise the required space linearly.

Flow utility is Stone-Geary over consumption and housing services, wrapped in CRRA:

$$u(c, \mathbf{h}; n, s) = \frac{\left[(c - \bar{c}(n, s))^{\alpha} (\mathbf{h} - \bar{h}(n, s))^{1-\alpha}\right]^{1-\sigma}}{1 - \sigma} + \psi_{\text{child}} m(n, s), \tag{38}$$

with utility equal to $-\infty$ when either residual is nonpositive. Housing services differ by tenure:

$$\mathbf{h} = \begin{cases} h^R, & d = R, \\ \chi_O H_k, & d = O_k, \end{cases} \qquad \chi_O > 0. \tag{39}$$

In market clearing, owners demand the physical quantity H_k, not the service quantity $\chi_O H_k$.

At death, households receive warm-glow bequest utility. The house passes to the estate at market value: there is no forced sale, no transaction cost, and no tax at death, so bequeathable wealth is $W^B = b + PH(d)$, liquid wealth plus the gross value of any house held. Bequest utility is

$$\mathcal{B}(W^B, n) = \theta_0(1 + \theta_n n)\frac{(\theta_1 + \max\{W^B, 0\})^{1-\sigma} - \theta_1^{1-\sigma}}{1-\sigma}. \tag{40}$$

The parameter θ_n lets completed fertility affect the desired estate even after children have left the household.

10 Housing, Finance, and Taxes

There is one aggregate housing-services market. The user cost q and asset price P satisfy

$$q = (r + \delta + \tau^p)P. \tag{41}$$

Housing supply is upward sloping:

$$H^S(q) = H_0\left(\frac{q}{\bar{q}}\right)^{\eta}. \tag{42}$$

The fixed-stock compact model is the limiting case $\eta = 0$ around the benchmark stock.

Renters choose c, h^R, b' subject to

$$c + qh^R + b' = Rb + y(a, z), \qquad 0 \le h^R \le \bar{h}^R, \qquad b' \ge 0, \tag{43}$$

where $R = 1 + r$. Owners of rung H_k choose c, b' subject to the owner branch budget and the borrowing limit described below. Owners pay maintenance and property taxes through $(\delta + \tau^p)PH_k$.

At origination, a buyer can finance at most share ϕ of house value, so the required down payment is

$$(1 - \phi)PH_k. \tag{44}$$

The post-purchase liquid position must satisfy

$$b' \ge -\phi PH_k. \tag{45}$$

An incumbent owner who keeps the same house is not forced to delever after a price change:

$$b' \ge \min\{b, -\phi PH_k\}. \tag{46}$$

This convention constrains origination but lets incumbency carry state dependence.

Housing adjustment is costly. A sale of house H_k incurs the proportional transaction cost ψ^s, so net sale proceeds are

$$\Omega(H_k) = (1 - \psi^s)PH_k. \tag{47}$$

Transaction costs are the only disposal friction in the computed benchmark: an owner who sells pays ψ^s per unit of value and faces no tax on accrued gains.

The capital-gains treatment of Part I is specified as an extension and is not part of the computed benchmark. Pricing realized gains requires adding a tax basis B^p to the owner's state; a sale would then trigger $\mathcal{T}^g(PH_k, B^p) = \tau^g \max\{PH_k - B^p - \bar{g}, 0\}$ with statutory exclusion $\bar{g}$, net proceeds would fall to $\Omega(H_k) - \mathcal{T}^g(PH_k, B^p)$, and stepped-up basis at death would forgive the accrued gain. The benchmark state vector (34) deliberately excludes B^p: the old-owner lock-in wedge $\bar{\ell}$ is priced analytically in Part I, and the counterfactual section instead bounds its quantitative content with a direct retention-margin experiment.

11 Fertility and Child Aging

Fertility is chosen during the fertile window

$$a \in \{A_f^{\text{start}}, \dots, A_f^{\text{end}}\}.$$

A childless household chooses

$$n' \in \mathcal{N} = \{0, 1, \dots, \bar{n}\}. \tag{48}$$

The option $n' = 0$ means no birth in the current period. A positive choice fixes completed fertility forever and moves the household into the first dependent-child state. A household that reaches the end of the fertile window without a positive choice remains childless.

Let $s = 0$ denote no dependent children and let $s = 1, \dots, S_c$ denote dependent-child stages. After a positive fertility choice, the household enters $s = 1$. Children age stochastically through the stages according to Π^c. While children are at home, $m(n, s) = n$; after maturity, $m(n, s) = 0$. Completed fertility still matters through bequest utility.

12 Household Problem

Within the period, the household first faces fertility if eligible, then chooses tenure and housing position, then chooses consumption, rental housing, and saving. Let the continuation value after income risk and child aging be

$$\bar{V}_{a+1}(b', d', z, n, s) = \sum_{z'} \sum_{s'} \Pi^z(z'|z) \Pi^c(s'|s, n) V_{a+1}(b', z', d', n, s'). \tag{49}$$

The terminal value is

$$V_{J+1}(b, z, d, n, s) = \mathcal{B}(W^B, n).$$

Conditional on renting, the branch value is

$$\begin{aligned} V_a^R(b, z, n, s) = \max_{c, h^R, b'} \quad & u(c, h^R; n, s) + \beta \bar{V}_{a+1}(b', R, z, n, s) \\ \text{s.t.} \quad & c + q h^R + b' = Rb + y(a, z), \\ & 0 \le h^R \le \bar{h}^R, \qquad b' \ge 0. \end{aligned} \tag{50}$$

Conditional on owning rung H_k after any adjustment, the branch value is

$$\begin{aligned} V_a^{O_k}(\tilde{b}, z, n, s) = \max_{c, b'} \quad & u(c, \chi_O H_k; n, s) + \beta \bar{V}_{a+1}(b', O_k, z, n, s) \\ \text{s.t.} \quad & c + (\delta + \tau^p) P H_k + b' = R\tilde{b} + y(a, z), \\ & b' \geq \underline{b}_k^O. \end{aligned} \tag{51}$$

The lower bound $\underline{b}_k^O$ is the origination or incumbent constraint in (45)–(46), depending on whether the household buys or keeps the current house.

Let $H(R) = 0$ and $H(O_k) = H_k$. The liquid wealth available after housing adjustment is

$$\tilde{b}(b, d, d') = b + \mathbf{1}\{d \in \mathcal{O},\ d' \neq d\}\Omega(H(d)) - \mathbf{1}\{d' \in \mathcal{O},\ d' \neq d\} P H(d'). \tag{52}$$

A seller collects net proceeds, a buyer pays the full purchase price, and a household that keeps its position pays nothing.

The tenure branch value for feasible d' is

$$\widetilde{V}_a^T(d'|b, z, d, n, s) = \begin{cases} V_a^R(\tilde{b}(b, d, R), z, n, s), & d' = R, \\ V_a^{O_k}(\tilde{b}(b, d, O_k), z, n, s), & d' = O_k. \end{cases} \tag{53}$$

Tenure and owner-rung choices have Type-I extreme-value shocks with scale κ_T:

$$\pi_a^T(d'|x) = \frac{\exp\{\widetilde{V}_a^T(d'|x)/\kappa_T\}}{\sum_{\hat{d} \in \mathcal{D}(x)} \exp\{\widetilde{V}_a^T(\hat{d}|x)/\kappa_T\}}, \tag{54}$$

and the inclusive tenure value is

$$V_a^T(x) = \kappa_T \log \sum_{\hat{d} \in \mathcal{D}(x)} \exp\{\widetilde{V}_a^T(\hat{d}|x)/\kappa_T\}. \tag{55}$$

For a childless household in the fertile window, the value of fertility option n' is

$$\widetilde{V}_a^F(n'|b, z, d) = V_a^T(b, z, d, n', s(n')), \qquad s(n') = \mathbf{1}\{n' > 0\}. \tag{56}$$

With Type-I extreme-value shocks of scale κ_F,

$$\pi_a^F(n'|b, z, d) = \frac{\exp\{\widetilde{V}_a^F(n'|b, z, d)/\kappa_F\}}{\sum_{\hat{n} \in \mathcal{N}} \exp\{\widetilde{V}_a^F(\hat{n}|b, z, d)/\kappa_F\}}. \tag{57}$$

Outside the fertile window, or after a positive fertility choice, this stage is degenerate and $V_a = V_a^T$.

13 Stationary Equilibrium

Let g be the stationary distribution of adult households over states. Adult mass is normalized to one for the benchmark quantitative exercise:

$$\int dg(x) = 1. \tag{58}$$

This normalization fixes the demographic envelope and lets fertility change household composition, tenure, and housing demand. It is not, by itself, a demographic closure for total births: the entry inflow of new adults is a fixed constant and births do not feed it. Quantitative counterfactuals therefore report per-capita objects — completed fertility, childlessness, tenure — at fixed population. The entry and scale margin of Part I is reinstated on top of the solved model by an outer fixed point described in Section 18.1, and its results are reported separately from the fixed-population results.

Definition 1 (Stationary equilibrium)**.** *A stationary equilibrium is a user cost and asset price* (q, P)*, value functions and policies, tenure and fertility probabilities, a stationary distribution* g*, and aggregate housing quantities* (H^D, H^S) *such that:*

1. *households solve the branch problems (50)–(51), tenure choice is given by (54), fertility choice is given by (57), and terminal utility is (40);*
2. *the distribution* g *is invariant under policies, income transitions, fertility shocks, child aging, deterministic aging, death, and entry of new adult households;*
3. *aggregate physical housing demand is*

$$H^D = \int \left[\mathbf{1}\{d = R\} h^R(x) + \sum_{k=1}^{K} \mathbf{1}\{d = O_k\} H_k \right] \mathrm{d}g(x); \tag{59}$$

4. *the housing market clears and the user-cost relation holds:*

$$H^D = H^S(q), \qquad q = (r + \delta + \tau^p)P; \tag{60}$$

5. *payroll taxes, property taxes, capital-gains taxes, pensions, and any rebates satisfy the government accounting rule specified for the calibration or counterfactual.*

14 Computation

The model is solved at an annualized four-year period. Households live $J = 17$ periods from age 18, retire at 66, and the last period in which a birth can occur begins at age 42. The wealth grid has $N_b = 120$ nodes on $[-35, 100]$ with power spacing concentrated near zero; income takes five persistent values $z \in \{0.6, 0.8, 1.0, 1.2, 1.4\}$ with per-period persistence 0.85 and invariant weights $(0.1, 0.2, 0.4, 0.2, 0.1)$; tenure is renting plus five owner rungs; fertility states are $n \in \{0, 1, 2+\}$ with one dependent-child stage. Value functions are computed by backward induction: the savings choice in each branch problem is solved by golden-section search, accelerated by Howard policy iteration; tenure and fertility follow the logit formulas of the model section. The stationary distribution follows by forward iteration of the exact transition operator, and the equilibrium price by damped fixed-point iteration on market clearing with a bracketing refinement on the scalar market, to tolerance 10^{-4}.

Because tenure and fertility choices are close to deterministic at the calibrated point, moments built on buy thresholds and the borrowing constraint move with the wealth grid, and grid resolution matters more here than in a standard lifecycle model. An earlier calibration at $N_b = 60$ re-evaluated 18 percent worse at $N_b = 120$ at the same parameters; part of its fit was grid noise. All searches in this paper therefore run at $N_b = 120$, and reported points are re-evaluated at $N_b = 240$; the fit section reports the result of that check.

15 Calibration

15.1 Externally set objects

Prices, finance, demography, and the housing menu are set outside the moment-matching loop.

Table 1: Externally set parameters

Object	Value	Source or rule
Interest rate r	4% per year	standard
Depreciation δ	2% per year	standard
Property tax τ^p	1% per year	U.S. average effective rate
Financed share ϕ	0.80	20% down payment
Sale cost ψ^s	0.06	transaction-cost evidence
Risk aversion σ	2	standard
Bequest shift θ_1	0.01	near-linear warm glow
Payroll tax τ_{pay}	0.179	pension balances in the stationary distribution
Income profile e_a	rises to 1.0 at 34, declines after 58	age-earnings profile
Owner menu $\{H_k\}$	$\{2, 4, 6, 8, 10\}$ rooms	room-count support, ACS
Renter cap $\bar{h}^R$	6.0 rooms	90th-percentile rental size rule of Greaney et al. (2025), measured in our matched ACS sample; 6.5 as robustness
Supply (H_0, η)	$(4, 1)$	unit-elastic benchmark
Entry wealth	PSID wealth-to-income ratios at age 22–25	first stage
Discount floor	$\beta_{\text{annual}} \geq 0.94$	declared restriction

The renter cap is the quantitative version of the compact model's $\bar{h}^R$. I measure it from the size distribution of rental units rather than estimate it. The calibrated fit is nearly flat within half a room of the measured value and deteriorates beyond, so the data cannot separate 6.0 from 6.5 but do reject anything further out. The discount floor binds in the estimation below, so β is in effect a restriction rather than an estimate.

15.2 Estimated parameters

Thirteen parameters are estimated by simulated method of moments: the discount factor β, the consumption share α, the adult floors $\bar{c}_0$ and $\bar{h}_0$, the child cost parameters $\bar{c}_n$, $\bar{h}_{\text{jump}}$, $\bar{h}_n$, and ψ_{child}, the choice scales κ_F and κ_T, the owner service multiplier χ_O, and the bequest parameters θ_0 and θ_n. The objective is a weighted sum of squared moment gaps, minimized by a global differential-evolution stage followed by self-reseeding local polish chains on a cluster, at $N_b = 120$.

Table 2: Estimates, search bounds, and bound status

Parameter	Estimate	Bounds	Status
β_{annual}	0.940	$[0.940, 0.995]$	at lower (declared restriction)
α	0.620	$[0.40, 0.95]$	interior
$\bar{c}_0$	1.279	$[0.08, 1.28]$	at upper
$\bar{c}_n$	0.116	$[0.05, 1.50]$	interior
$\bar{h}_0$	1.000	$[1.0, 6.0]$	at lower
$\bar{h}_{\text{jump}}$	2.120	$[0.05, 2.50]$	interior, high
$\bar{h}_n$	1.382	$[0.02, 2.00]$	interior
ψ_{child}	0.349	$[0, 0.35]$	at upper
κ_F	1.019	$[1.0, 12.0]$	at lower
κ_T	0.000	$[0, 0.12]$	at lower
χ_O	1.150	$[0.40, 1.15]$	at upper
θ_0	0.0015	$[0, 2.0]$	at lower
θ_n	0.972	$[0, 1.50]$	interior

Eight of thirteen parameters sit at or against the search box. The data push these parameters to the boundary of the admissible space, and the structural misses documented below are the same tension seen from the moment side.

The two choice scales land at corners with different meanings. The tenure scale κ_T estimates to zero: the data prefer deterministic tenure choice, and the smoothing device does no work. The fertility scale κ_F estimates to its floor, where the birth choice is also near-deterministic, but there the choice structure carries the entire fertility block. At a near-degenerate logit, low-income types have birth probabilities of exactly zero, and fertility moments inherit sensitivity to the discreteness of the wealth grid. The grid protocol of Section 14 guards against the discreteness; whether taste smoothing belongs in these choices at all is a question the estimation cannot settle.

15.3 Target moments and fit

The calibration targets fourteen moments: completed fertility and childlessness from the CPS (the June 2024 supplement cells for women 40–44), tenure and housing-size moments by age and family status from a matched ACS metropolitan sample, and wealth moments from the PSID. The table reports every target with its model moment, gap, weight, and loss contribution. The total loss at the reported point is 6.31.

Table 3: Target moments and model fit ($N_b = 120$; loss 6.31)

Moment	Target	Model	Gap	Weight	Contribution
Old-age ownership rate (65–75)	0.764	0.893	+0.128	160	2.64
Ownership rate 25–34	0.341	0.226	−0.115	80	1.05
Owner−renter mean rooms (childless 30–55)	2.419	2.151	−0.268	12	0.86
Family ownership gap	0.168	0.298	+0.130	45	0.77
Young renter liquid wealth / income (25–35)	0.179	0.353	+0.173	12	0.36
Aggregate ownership rate	0.575	0.531	−0.045	100	0.20
Childless renter mean rooms (30–55)	3.805	3.655	−0.150	6	0.14
Childlessness rate	0.188	0.260	+0.072	20	0.10
Rooms gap, 3+ vs. 1–2 children (30–55)	0.368	0.269	−0.099	8	0.08
Old parent−childless nonhousing wealth gap	1.007	0.842	−0.166	2	0.05
Housing increment at first birth (rooms)	0.664	0.618	−0.046	14	0.03
Completed fertility	1.918	1.949	+0.031	20	0.02
Old nonhousing wealth / income, median (65–75)	2.230	2.328	+0.097	0.8	0.01
Childless owner share, rooms ≥ 6	0.596	0.593	−0.003	25	0.00

The model matches the fertility level, the size-distribution moments, and the housing response to a first birth. Three misses are structural, and each has a diagnosis. Old-age ownership is too high: the model keeps 89 percent of 65–75 households in owned housing against 76 percent in the data, because nothing in the model pushes an old owner out of her house. There are no health shocks and no mortality-linked downsizing, so retention is close to absorbing. Ownership at 25–34 is too low (0.23 against 0.34) while young renter wealth is too high (0.35 against 0.18); these are two faces of one timing failure. Model households delay births toward the fertility deadline and buy at the birth, where the data show earlier purchases. And childlessness overshoots (0.26 against 0.19), which is what pricing children through goods and space floors alone does to the extensive margin.

Re-evaluated at $N_b = 240$ at the same parameters, the loss rises from 6.31 to 6.84, and the drift concentrates in the family ownership gap and the old-age ownership rate, both of which move further from their targets at the finer grid. The table's numbers are therefore, if anything, generous to the model on its two main misses, and the diagnosis does not depend on the grid.

16 The Baseline Equilibrium

Before any counterfactual, it is worth seeing what the stationary equilibrium looks like. The figures below show policies, tenure, fertility, housing, wealth, and market clearing.

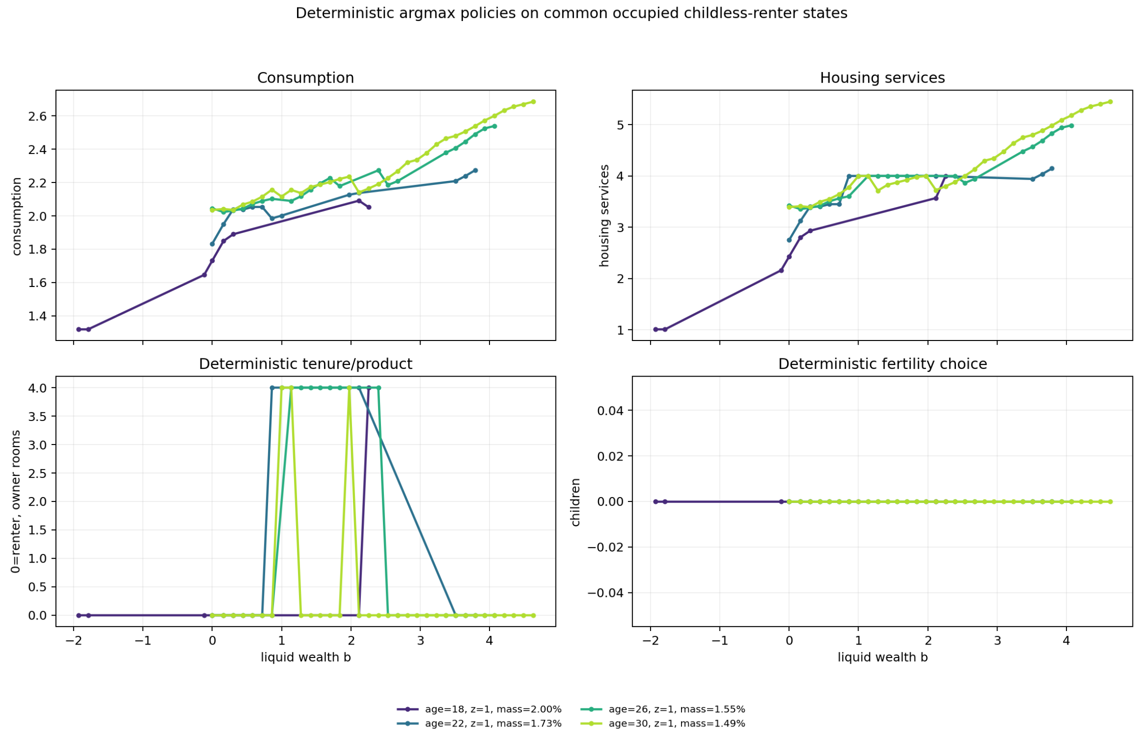

Figure 2: Policy functions over wealth by age and family state. Savings and housing policies are monotone in wealth; the kinks are the down-payment thresholds at the owner rungs.

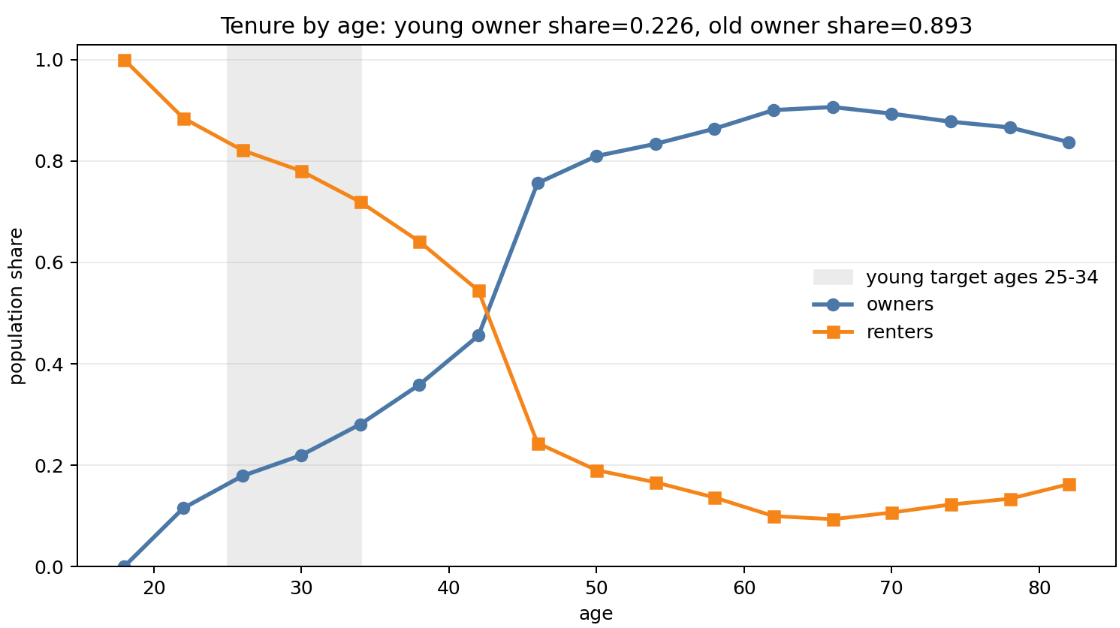

Figure 3: Tenure over the lifecycle. Ownership rises steeply at the end of the fertile window and stays high into old age, the two ends of the intergenerational allocation problem.

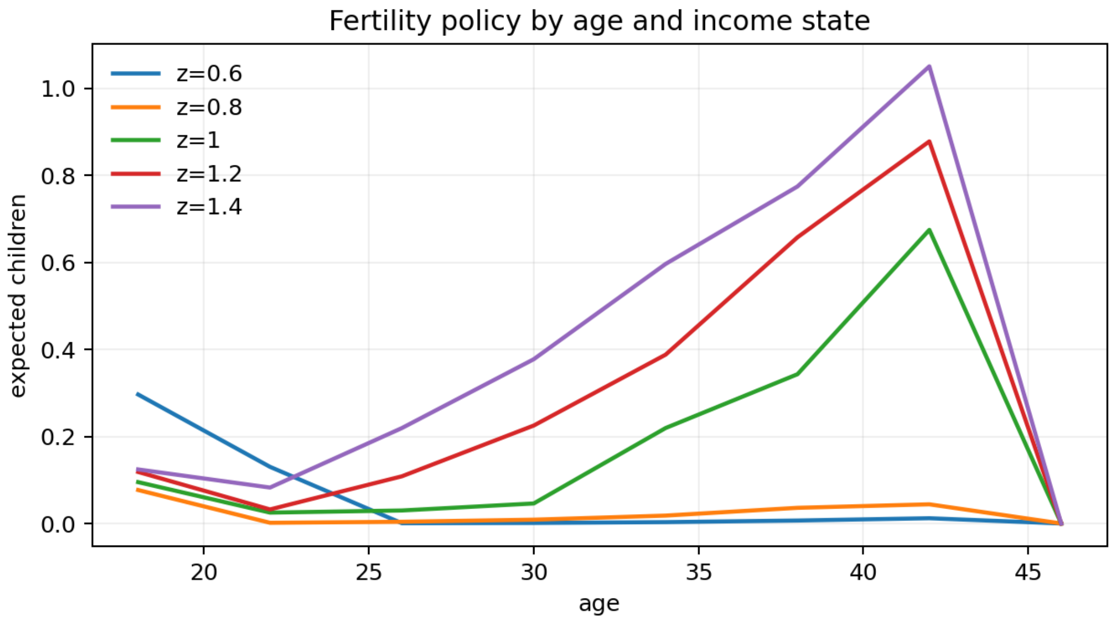

Figure 4: Birth probabilities by wealth and income state. At the calibrated near-deterministic choice scale, low-income types have birth probabilities of exactly zero.

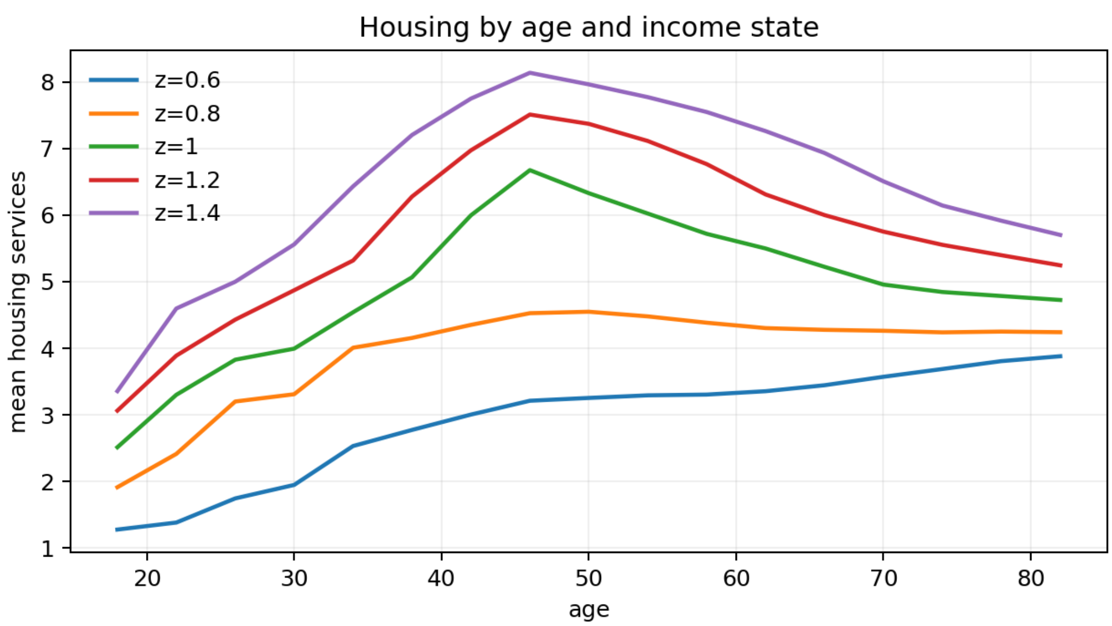

Figure 5: Housing services by age, tenure, and family state. Renters with children mass at the rental cap; owners occupy the discrete rungs.

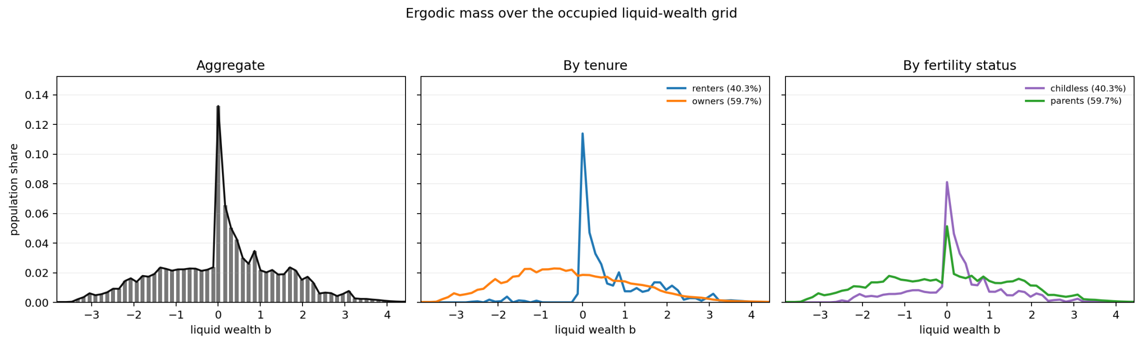

Figure 6: Wealth accumulation over the lifecycle. Young renters concentrate near zero liquid wealth; owners lever into negative liquid positions at purchase.

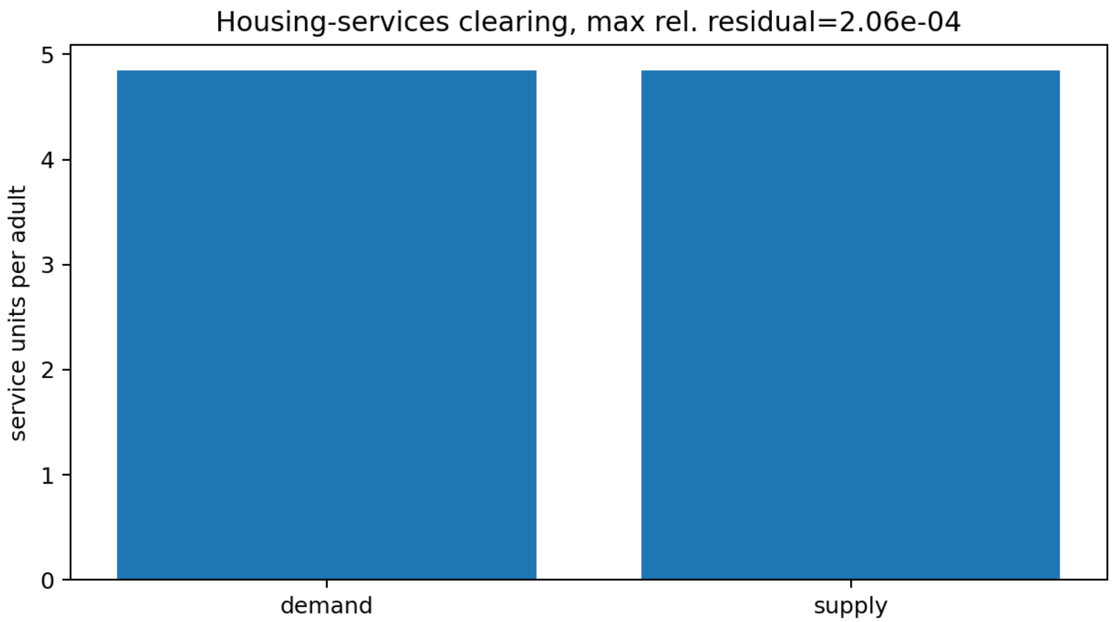

Figure 7: Market clearing. Aggregate demand against the unit-elastic supply curve at the equilibrium user cost.

Renting households with children sit against the rental cap. Family-sized space is the scarce object, and it is scarce as availability rather than price: within a size cell, the model rent carries no premium. Ownership arrives late, in a jump at the end of the fertile window. The next section shows that the jump is birth-driven purchasing at the deadline, not gradual saving toward a threshold.

17 The Mechanism Quantified

Part I's argument runs through one object, the access gap ζ: the amount by which a constrained household's marginal valuation of family space exceeds its market user cost. Under equal scaling, the corollary to Proposition 2 says the same gap prices misallocation, fertility exposure, and entry.

This section measures it in the calibrated equilibrium.

The gap is large, and it covers most of the fertility margin. Among fertile-age renting households pinned at the rental cap, the mean access gap is 0.248 goods units, 1.28 times the market user cost, so capped renters value a marginal room at 2.3 times what the market charges for it. Weighting fertility responses by exposure as in Proposition 2, 73 percent of the model's fertility-response mass sits on households with $\zeta > 0$. The share is stable across grid resolutions and across alternative owner menus. What generates it is the geometry between the rental cap and the family space floor, not the placement of the owner rungs.

Ownership arrives at the fertility deadline. The lifecycle jump in Figure 3 is concentrated in the last fertile period: the buy region expands exactly when the option value of waiting to have children expires, and buyers arrive at the threshold with wealth well above the required down payment. Housing access does not gate births by forcing households to save toward a distant constraint. It compresses births and purchases into a joint decision at the deadline, and this is why relaxing the down-payment constraint alone creates few births in the counterfactuals below. The binding failure is timing, not exposure.

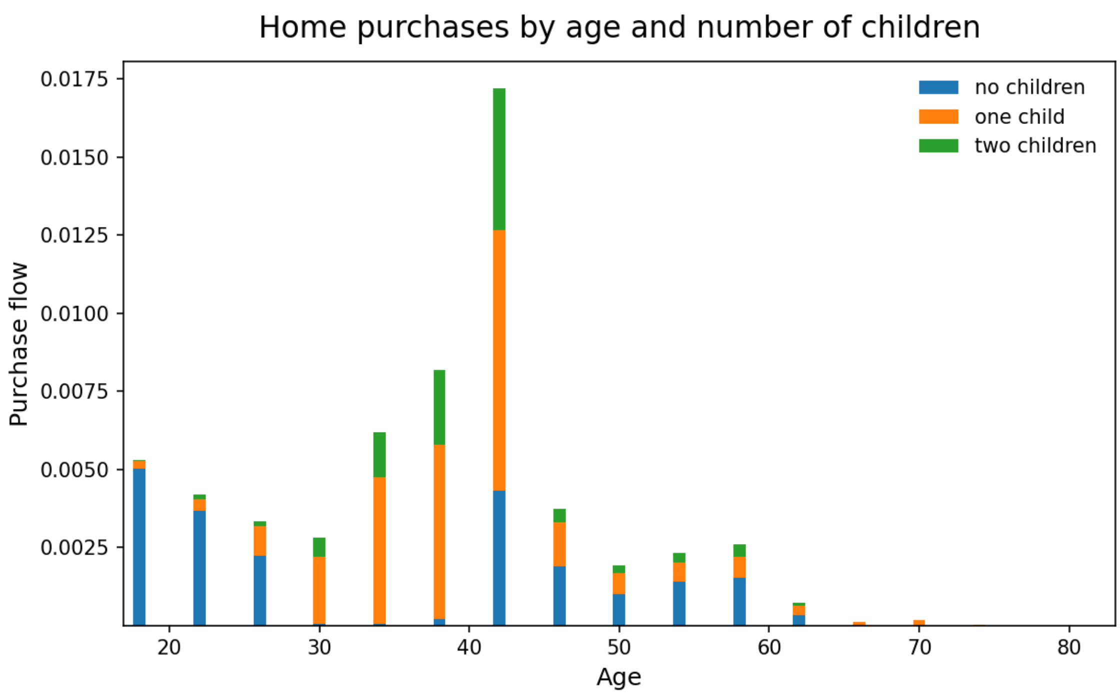

Figure 8: Purchase flows by age and family state. Buying is concentrated at the end of the fertile window and is birth-linked.

The compact model's misallocation proposition is an accounting statement about who holds family-capable space, and it can be checked against the data. The table reports the allocation of the owner stock with six or more rooms, in the calibrated equilibrium and in the matched ACS sample.

Table 4: Who holds the family-sized owner stock (rooms ≥ 6)

Holder	Data (ACS)	Model
Age 18–29	3.4%	0.4%
Age 30–45	24.8%	12.7%
Age 46–64	43.3%	50.5%
Age 65+	28.5%	36.5%
Held by households 46+	71.8%	87.0%
Held by households with no children at home	54.9%	72.5%

The concentration is in the data, not only in the model. In the ACS, 72 percent of the family-sized owner stock is held by post-fertile households and 55 percent by households with no children at home. The model reproduces the pattern and overstates it, in the directions of its two diagnosed misses: late births push fertile-age holdings down, and the missing old-age exit margin pushes old holdings up. The households the planner comparison wants served, fertile-age with positive access gaps, hold 13 percent of the family-sized stock in the model, and two-thirds of fertile-age parents do not own family-sized space at all.

The old side of Part I's surplus formula has a direct quantitative counterpart in a retention-margin experiment. Imposing a disposal wedge on old owners of large homes moves prices by less than one percent and completed fertility by less than a hundredth. This is not evidence that the old side of the misallocation is small. The calibrated old are at a retain-everything corner: with no health, mortality, or downsizing margin, essentially no old household is marginal to the wedge, and a tax on a margin nobody uses is free. The experiment bounds the planner surplus from below. The stock table measures the allocation directly; an old-age exit margin is the extension that would let prices reveal its value.

18 Policy Counterfactuals

The instruments follow the access margins of the compact model: a holding tax on large owner units (the capitalization channel of eq. 31), birth-linked down-payment grants restricted to family-sized rungs (the financing gap $\zeta^{O,F}$), variation in the rental cap $\bar{h}^R$, a retention wedge on old owners, and a revenue-linked package combining the tax and the grant. All results in this subsection hold population fixed; prices re-clear in every experiment.

The tax capitalizes but does not create births. A two percent holding tax on large units lowers their asset price by 12 percent, and the capitalization is invariant to grid resolution, moment weights, and the owner menu. Completed fertility rises by only +0.004 to +0.009. The pass-through works exactly as eq. 31 says, and down payments do fall. But exposed households buy at the fertility deadline with wealth already above the threshold, so cheaper access changes which house they buy more than whether they have children.

Grants create births, with a dose–response. A down-payment grant paid at a birth, restricted to family-sized rungs, raises completed fertility by +0.01, +0.13, +0.26, and +0.49 at grant sizes 0.1, 0.4, 0.58, and 1.0, in units of mean annual income. The access instrument works where the price instrument does not because it is conditional on the birth itself: it closes the financing gap at the moment the household is on its fertility margin. The conditioning also distorts. Small grants delay pre-birth purchases, since households near a birth wait for the subsidy, and young ownership falls from 0.23 to 0.09 at the smallest grant. The distortion fades at larger grant sizes, but any conditional instrument inherits the margin.

The package is the decentralized version of Part I's planner reallocation. The two percent tax on large homes funds the birth-linked grant; the tax lowers incumbents' retention value and the price of large units by 11 percent, and the grant closes the young side's financing gap at the birth margin. At the reported calibration the package raises completed fertility by +0.15 and is revenue-funded.

18.1 The entry and scale margin

The results above hold population fixed. The entry margin of Part I is reinstated by an outer fixed point around the solved model: the policy changes household values, values move the entry probability against the outside option and the number of city-born future entrants, the implied stationary population scale S multiplies aggregate housing demand, and the price re-clears against the supply curve. Households never see S directly; population size operates through the price of space. The outside value and entry scale are recovered at the benchmark so that the baseline is exactly $S = 1$. The recovered conventions are documented rather than estimated, and the binding one is the assumption that city-born children become future entrants one-for-one.

These results are computed at the working grid ($N_b = 120$) at the headline calibration; the fixed-population column re-solves each case inside the same protocol, so the comparison is internal. The grant and package rows carry a few-percent price imprecision from the outer loop's fast solves; their directions and orders of magnitude are robust.

Table 5: Counterfactuals with the entry margin on ($N_b = 120$)

Policy	ΔTFR, fixed pop.	ΔTFR, intensive	Scale S	Total births
Holding tax 2%	+0.004	+0.002	1.003	+0.4%
Grant 0.4, rungs ≥ 6	+0.119	+0.083	1.038	+8.2%
Package (tax + grant)	+0.162	+0.086	1.040	+8.6%
Rental cap 6.5	−0.006	−0.009	1.004	−0.0%
Retention wedge 0.2	+0.006	+0.000	1.007	+0.7%

The grant pulls in scale, scale bids up prices, and the capitalization nearly triples with entry on; about a third of the fixed-population fertility gain is given back, while total births rise by eight percent. Per-capita fertility overstates what an access policy does to a place, and total births understate what it does per family. The scale response is a demographic feedback rather than an entry-elasticity artifact. Across policies the entry probability barely moves, and city-born future entrants do the work, which is why the results survive changes in the entry conventions the data do not pin down: halving the assumed entry probability moves the tax's births effect from +0.40 to +0.46 percent. The tax creates almost no births at fixed population and almost none with entry on, and the retention wedge, a tax on a downsizing margin the calibrated old do not use, is near-neutral everywhere. Whether local birth gains count as births created or births relocated is a question of city versus national accounting, and any local-policy conclusion has to answer it explicitly.

19 Limitations

Each of the calibration's weaknesses names a missing margin. Old-age ownership is too high because nothing lets an old owner exit; a health or downsizing margin is the natural extension. The young ownership-and-wealth block fails through fertility timing: in the model, delaying a birth costs nothing, so births and purchases pile up at the deadline. The extension that would repair this is a mechanism that makes delay itself costly, such as fecundity that declines with age. The same mechanism bears on the childlessness overshoot, since households that wait and fail add involuntary childlessness of the kind goods-and-space child costs cannot generate.

Three further limitations are matters of scope. The computed benchmark prices old-owner lock-in through transaction costs only; the capital-gains and basis machinery of Part I is specified but not computed, so the retention experiment bounds rather than measures that wedge. The quantitative supply curve is unit-elastic while the compact model's misallocation statement is cast at a fixed stock; how much of the tax capitalization result depends on the supply elasticity is an open sensitivity, and a near-fixed-stock solve would bound it. And roughly a quarter of the entrant wealth distribution injected from the PSID first stage sits at states infeasible for the model's consumption floor, which interacts with any conclusion about the young wealth block.

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